Five-gluing novelty audit: all selected source classes overlap history

2026-10-25

All 12 source trace/isogeny classes selected by the new five-torsion screen were already present in the retained genus-three sweep. No new local source isogeny class was found.

This is a supplement to the census and proof report. It resolves the local-history question left open by that snapshot. The module classification and sufficient geometric criterion remain derived written results; their publication priority remains unresolved. The user's requested novel affected family has not been established.

Read-only historical comparison

The completed exact source census selected 252 full-degree F512/F8 models, forming 126 twist pairs and 12 source trace classes. We compared those exact traces with saved genus-three, base-degree-three candidates in the pre-existing `runs/smoke.sqlite`, retaining only raw-verified cover models and independently verified source orders. The lookup scope was frozen at `dee8ef6e` before the bulk comparison. No search job or timing campaign was run.

Source trace	Verified historical cover-model matches
-43	3
-37	9
-23	36
-17	30
-13	27
-7	45
7	45
13	27
17	30
23	36
37	9
43	3

These counts enumerate historical cover-model rows, not independent elliptic families or successful attacks. Each trace has a retained witness containing its exact spectral model ID, model JSON, source JSON and verification flags. The database SHA-256 is `8290f8fbb6abc6c2e16e8ce30484c45d979c785e2098d4332f1e1478e621bd1c`. The exact SQL and all witnesses are saved in `cubic-gluing-checks/five-only-historical-overlap.json`.

What the overlap establishes

For elliptic curves over the same finite field, equality of trace identifies the same rational isogeny class. It does not establish equality of the exact coefficient b or a checked map between the saved source representative and our new source model. The old records are spectral/source matches and retain `explicit_map_to_target_verified=false`. They supply no explicit old transfer to substitute for the missing new quotient map.

The five-torsion predicate selects models excluded by the earlier three-torsion predicate. Its source trace classes are nevertheless already in the broader historical genus-three screen. A different recognition rule, an alternate

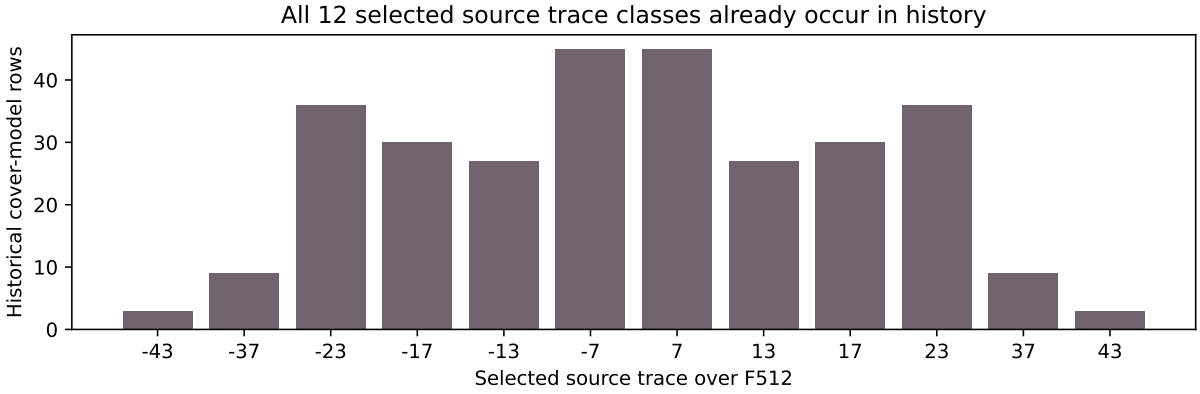

Figure 1: Verified historical matches for every selected trace. This is an exact journal lookup, with no sampling uncertainty or large-field extrapolation.

quotient and a new local source class are separate statements. Only the recognition rule and the stated existence implication have been derived here. No previously unknown vulnerable family, useful cost advantage, or evaluated five-gluing attack is claimed.

The full Tian 2023 primary-text comparison remains unfinished, and Hess's generalized descent remains the broad binary baseline. The next construction obligations on b=48 are actual E[5] coordinates, quotient reconstruction, arithmetic twist selection and an evaluated prime-subgroup/log map. Those steps could establish a different explicit correspondence; they do not erase this historical source-class overlap.

The previous 89-file manifest and all earlier reports remain unchanged. This supplement has its own additional-artifact manifest. The original curve catalogue and verified attack-route graph receive no new entry from this result.

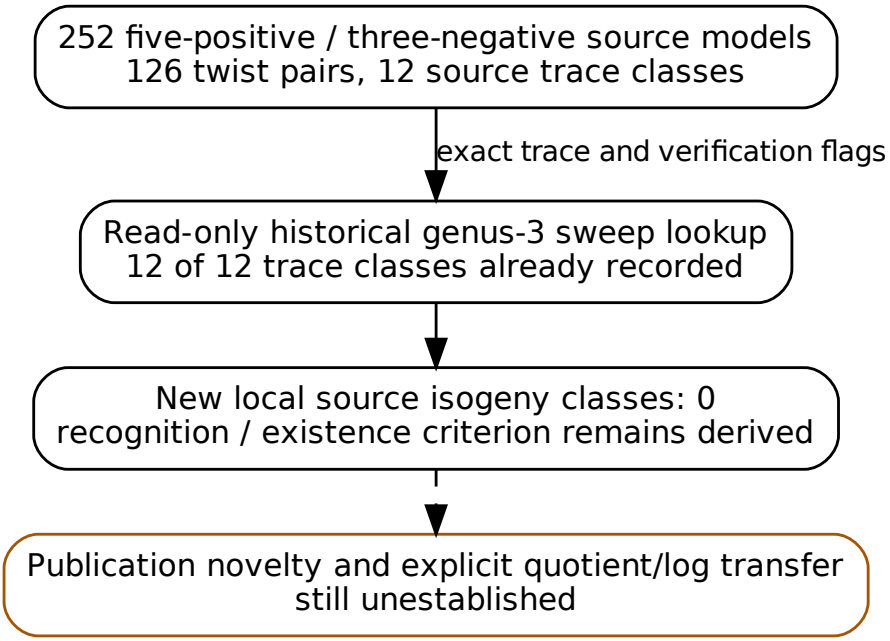

Figure 2: Evidence boundary after the historical lookup.